Risk Structure in CDS Spreads: A Multivariate Approach

STA437 Final Project

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1 Introduction

This project analyzes CDS spread data for over 600 companies across ten maturities (PX1–PX10). These spreads reflect the market's perception of each company's credit risk.

Our goal is to understand how companies differ in their credit risk profiles and whether there are underlying patterns in the CDS term structure. We explore three questions:

- Can a small number of components explain most of the variation in CDS spreads?
- Can we group companies into meaningful credit risk clusters?
- Are there latent factors driving the relationships among CDS maturities?

To answer these, we apply: - **PCA** to reduce the dimensionality of the 10-tenor data. - **K-means clustering** on the PCA scores to identify company risk groups. - **Factor Analysis** to interpret the relationships among maturities.

2 Data and Preprocessing

The dataset includes daily CDS spreads for over 1,000,000 observations across more than 600 companies. Each company has CDS quotes for ten maturities, labeled PX1 through PX10.

To reduce noise and focus on firm-level credit risk, we aggregated the data by computing the average CDS spread for each company across time. This resulted in a dataset of 661 companies, each represented by ten features (PX1–PX10).

We then standardized the PX columns and examined the distribution of CDS spreads across tenors to ensure data quality before continuing with multivariate methods.

2.1 Load the CSV

```r
suppressPackageStartupMessages({
  library(dplyr)
  library(kableExtra)
})

cds_data <- read.csv("~/STA437/final project/cds_data.csv")

head(cds_data) %>%
  kbl(caption = "Sample of CDS Spread Data (without Ticker)", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
                position = "center")
```

Table 1: Sample of CDS Spread Data (without Ticker)

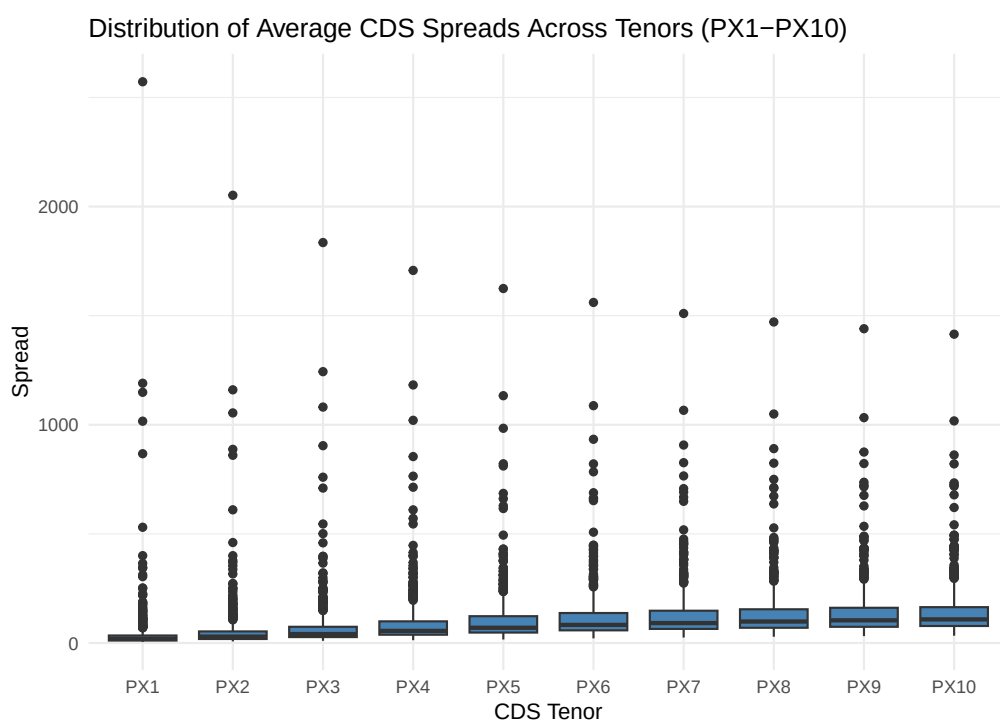
Date	Ticker	Company	PX1	PX2	PX3	PX4	PX5	PX6	PX7	PX8	PX9	PX10
2015-01-01	A	Agilent Technologies Inc	14.850	25.05	39.055	73.715	86.30	111.305	125.905	144.300	158.600	168.195
2015-01-02	A	Agilent Technologies Inc	14.855	25.15	39.160	73.600	86.40	111.510	126.100	144.375	158.775	168.375
2015-01-05	A	Agilent Technologies Inc	13.455	24.96	39.310	70.200	88.26	111.970	125.060	143.165	157.385	166.825
2015-01-06	A	Agilent Technologies Inc	15.535	24.88	39.300	72.680	91.10	116.610	130.890	150.365	165.655	175.885
2015-01-07	A	Agilent Technologies Inc	15.215	25.50	40.270	73.880	92.50	117.780	131.940	151.145	166.215	176.295
2015-01-08	A	Agilent Technologies Inc	14.765	26.10	40.620	73.640	93.08	116.620	129.470	148.505	163.365	173.275

2.2 Preprocessing

```
library(dplyr)
cds_avg <- cds_data %>%
  group_by(Ticker, Company) %>%
  summarise(across(starts_with("PX"), \(x) mean(x, na.rm = TRUE)), .groups = "drop")
```

2.3 Visualize CDS Term Structure

```
library(reshape2)
library(ggplot2)

cds_long <- melt(cds_avg[, -1], id.vars = "Company")

ggplot(cds_long, aes(x = variable, y = value)) +
  geom_boxplot(fill = "steelblue") +
  labs(title = "Distribution of Average CDS Spreads Across Tenors (PX1-PX10)",
     x = "CDS Tenor", y = "Spread") +
    theme_minimal()
```



The boxplot shows that CDS spreads generally increase with maturity, which is consistent with typical

term structure behavior: longer-term contracts carry more uncertainty and higher risk premiums. The distributions are fairly stable and there are no missing values, confirming the data is well-prepared for PCA and clustering.

3 Methodology

We used the following multivariate techniques:

- **Principal Component Analysis (PCA)** to reduce the 10-dimensional PX1–PX10 spread data to 2 components, retaining over 99% of the variance. This enabled low-dimensional visualization and simplified analysis.
- **K-means clustering** on the PCA results to identify groups of companies with similar credit spread profiles. We used the elbow method to choose the optimal number of clusters.
- **Factor Analysis** with varimax rotation to uncover latent structures driving the correlation among CDS maturities. This helped interpret the underlying dimensions of risk across short and long maturities.

These methods were chosen for their ability to reduce complexity and reveal hidden structure in multivariate financial data.

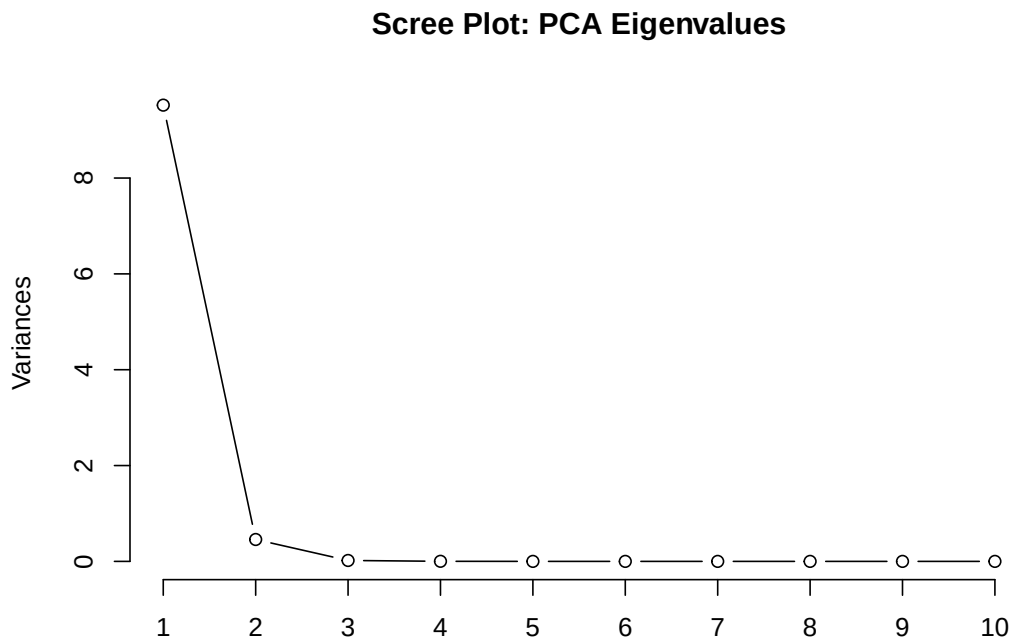
4 Results

4.1 Principal Component Analysis (PCA)

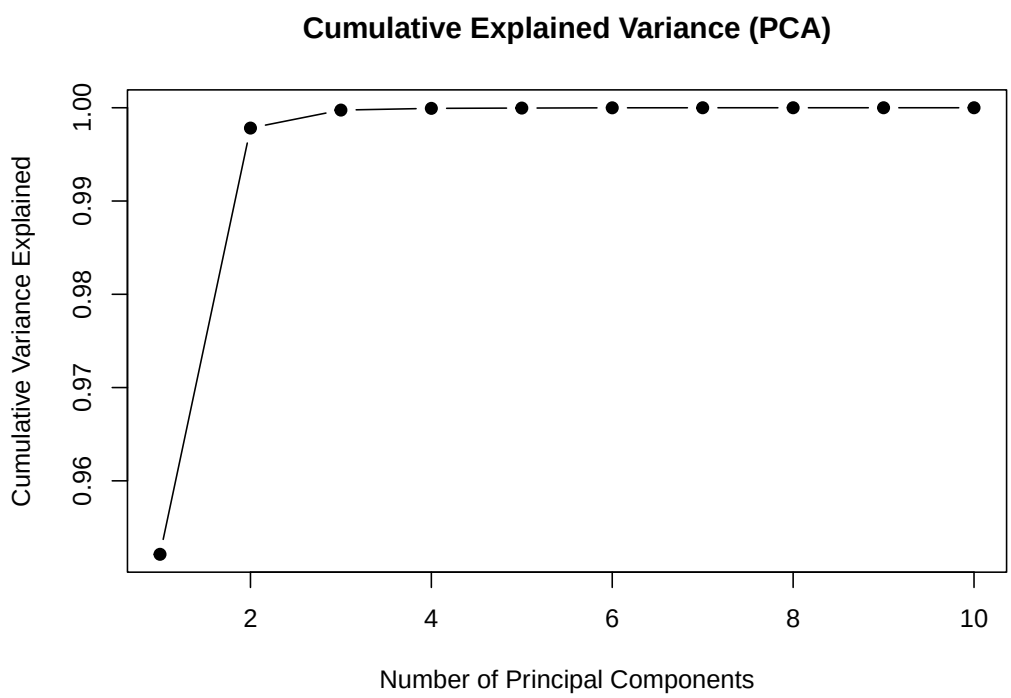
To reduce the dimensionality of the ten CDS spread variables (PX1–PX10), we applied Principal Component Analysis (PCA) after standardizing the data. PCA helps identify the dominant components driving variance in the dataset, allowing us to summarize the spread structure and visualize companies in a low-dimensional space.

4.1.1 Scree Plot and Cumulative Variance

```r
cds_matrix <- cds_avg[, 3:12]  # PX1 to PX10

# Standardize
cds_scaled <- scale(cds_matrix)

# PCA
pca_result <- prcomp(cds_scaled, center = TRUE, scale. = TRUE)

# Scree plot (eigenvalues)
plot(pca_result, type = "lines", main = "Scree Plot: PCA Eigenvalues")
```




```r
# Cumulative variance
cum_var <- cumsum(pca_result$sdev^2 / sum(pca_result$sdev^2))
plot(cum_var, type = "b", pch = 19,
     xlab = "Number of Principal Components",
     ylab = "Cumulative Variance Explained",
     main = "Cumulative Explained Variance (PCA)")
```



4.1.2 PCA Projection of Companies

```r
pca_df <- cds_avg[, c("Ticker", "Company")]
pca_df$PC1 <- pca_result$x[,1]
```

Table 2: PCA Scores for First 6 Companies

Ticker	Company	PC1	PC2
A	Agilent Technologies Inc	-0.0115491	0.2958879
AALLN	Anglo American PLC	2.8146634	1.0240691
AAPL	Apple Inc	-1.7276306	-0.3491678
ABBNVX	ABB Ltd	-2.0242358	-0.6026961
ABBV	AbbVie Inc	-1.0238001	-0.2555187
ABC	AmerisourceBergen Corp	-0.9457271	-0.0508519

```
pca_df$PC2 <- pca_result$x[,2]
library(kableExtra)

head(pca_df) %>%
  kbl(caption = "PCA Scores for First 6 Companies") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "bordered"),
                full_width = FALSE,
                position = "center")
```

The scree plot shows a strong drop after the first component, and the cumulative variance plot confirms that the first principal component alone explains 95.2% of the variance, while the first two together explain 99.8%. This means that most of the variation in CDS spreads across maturities can be captured in just two dimensions.

The PCA projection shows companies spread along PC1 and PC2, indicating that PC1 likely represents a general credit risk level (across all maturities), while PC2 captures more subtle differences, possibly in curve shape. These results justify reducing the data to two components and proceeding with clustering in the next section.

4.2 Clustering

With the PCA-reduced data, we apply K-means clustering to group companies based on their CDS spread profiles. Since the first two principal components explain almost all the variation in the data, we perform clustering in this two-dimensional space. This allows us to identify distinct credit risk patterns among firms.

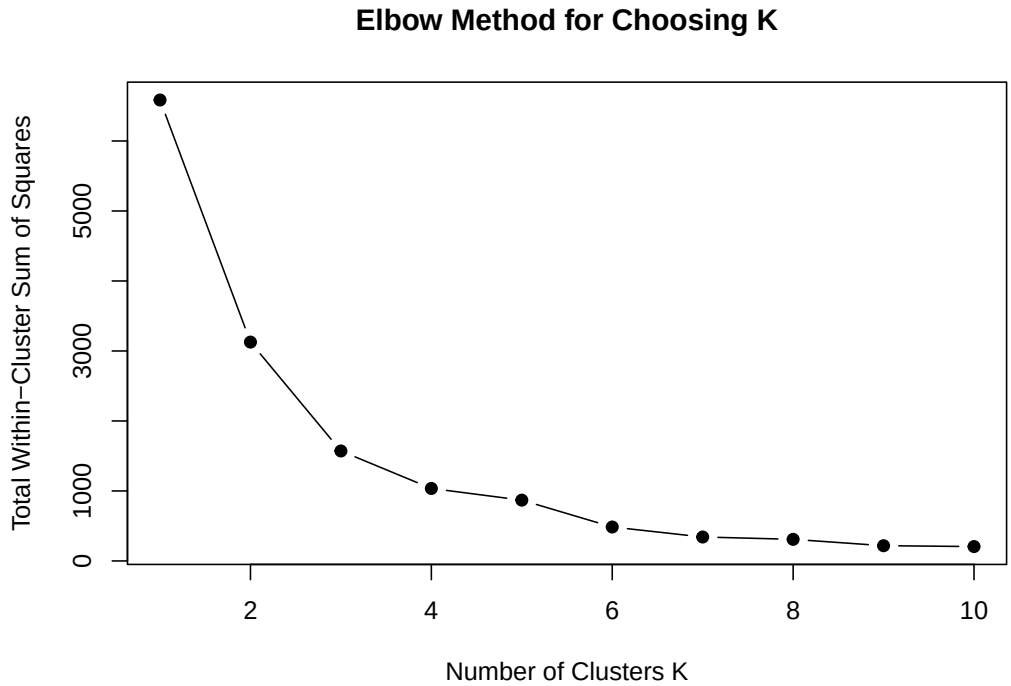
4.2.1 K-Means Clustering

We begin by selecting the number of clusters using the elbow method, then fit the K-means model and visualize the clusters.

4.2.2 Determine number of clusters (k) using the Elbow Method

```
# Code for computing and plotting elbow method (WSS)

set.seed(123)  # for reproducibility

wss <- sapply(1:10, function(k){
  kmeans(pca_df[, c("PC1", "PC2")], centers = k, nstart = 10)$tot.withinss
})

plot(1:10, wss, type = "b", pch = 19,
     xlab = "Number of Clusters K",
```

```
     ylab = "Total Within-Cluster Sum of Squares",
     main = "Elbow Method for Choosing K")
```

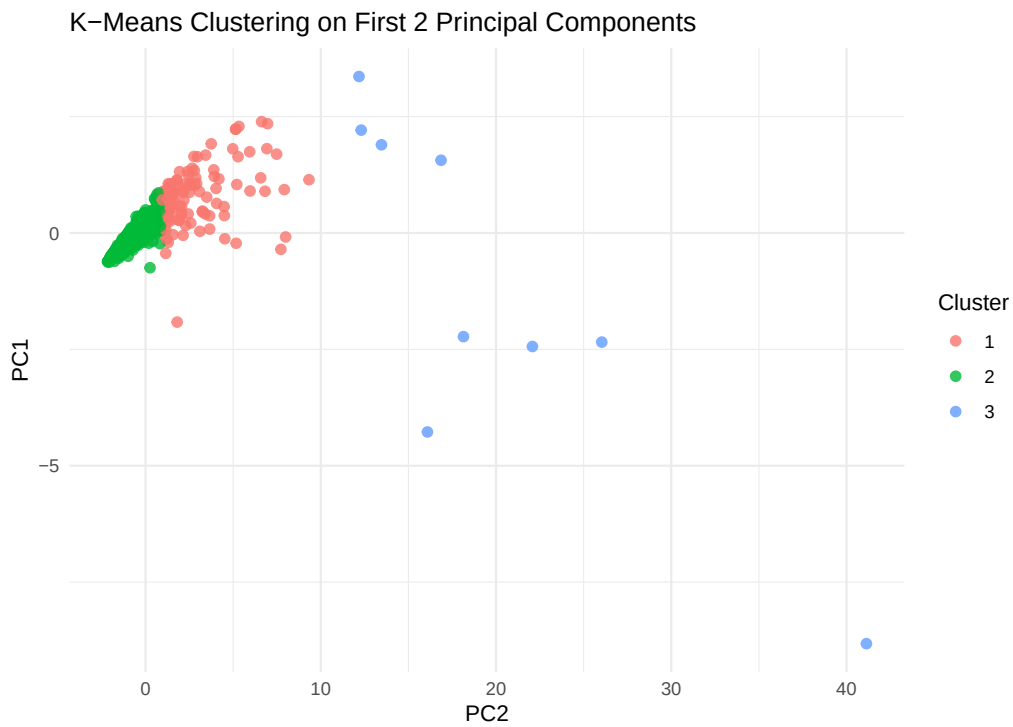


4.2.3 Perform K-means with chosen k (try k = 3)

```
# Code to fit K-means with k = 3 and assign clusters
kmeans_result <- kmeans(pca_df[, c("PC1", "PC2")], centers = 3, nstart = 25)
pca_df$Cluster <- as.factor(kmeans_result$cluster)
```

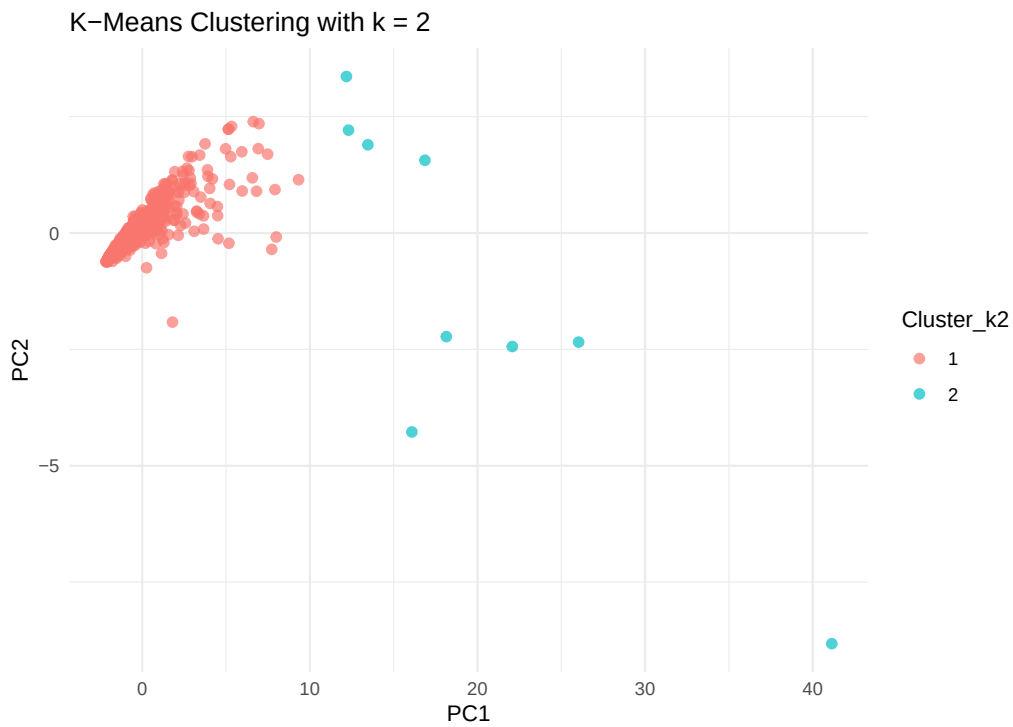
4.2.4 Visualize clusters

```
# Code for ggplot cluster scatter plot (PC1 vs PC2)

library(ggplot2)

ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster)) +
  geom_point(size = 2, alpha = 0.8) +
  labs(title = "K-Means Clustering on First 2 Principal Components",
       x = "PC2", y = "PC1") +
  theme_minimal()
```

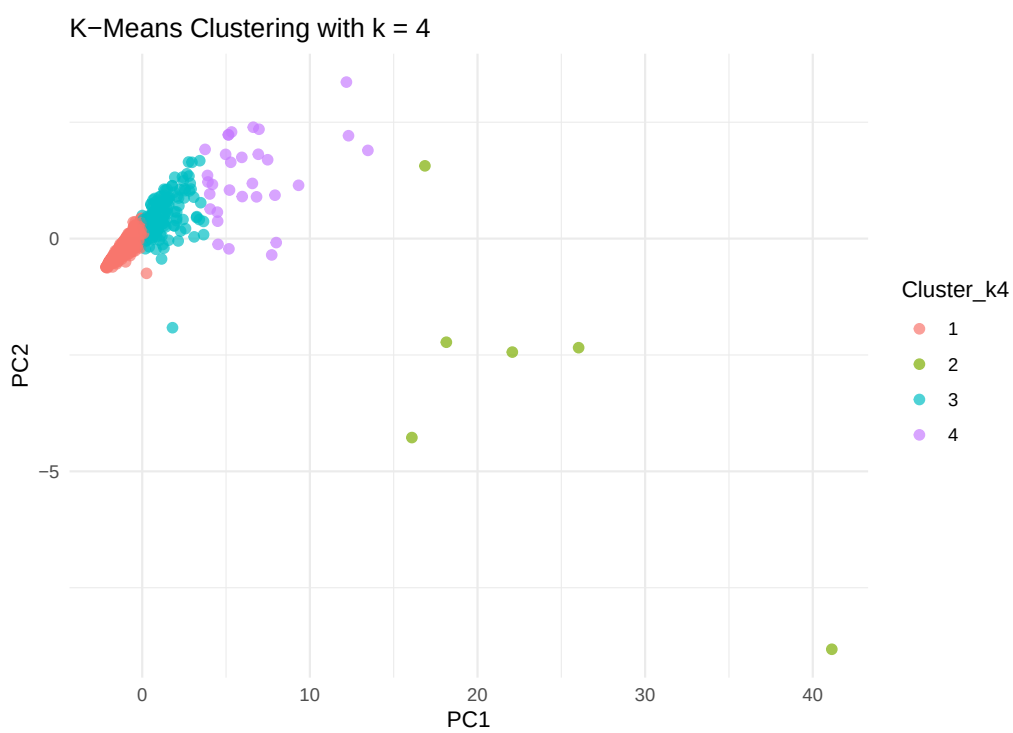


4.2.5 Try Different k Values

```
# Code to plot for k = 2 and k = 4

# Try k = 2
set.seed(123)
kmeans_2 <- kmeans(pca_df[, c("PC1", "PC2")], centers = 2, nstart = 25)
pca_df$Cluster_k2 <- as.factor(kmeans_2$cluster)

# Try k = 4
set.seed(123)
kmeans_4 <- kmeans(pca_df[, c("PC1", "PC2")], centers = 4, nstart = 25)
pca_df$Cluster_k4 <- as.factor(kmeans_4$cluster)


#### plot for k=2
  ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster_k2)) +
  geom_point(size = 2, alpha = 0.7) +
  labs(title = "K-Means Clustering with k = 2",
       x = "PC1", y = "PC2") +
  theme_minimal() +
  theme(legend.position = "right")
```




```r
#### plot for k=4
  ggplot(pca_df, aes(x = PC1, y = PC2, color = Cluster_k4)) +
  geom_point(size = 2, alpha = 0.7) +
  labs(title = "K-Means Clustering with k = 4",
       x = "PC1", y = "PC2") +
  theme_minimal() +
  theme(legend.position = "right")
```



The elbow plot shows a clear bend at k = 3, indicating a natural grouping at this level. The cluster plot based on PC1 and PC2 confirms that the data separates cleanly into three groups. Visual comparison with k = 2 and k = 4 shows that k = 2 is too coarse and k = 4 begins to overfit by splitting naturally cohesive groups.

This justifies selecting k = 3 as our final clustering solution and sets the stage for interpreting what these groups represent in terms of credit risk.

4.3 Cluster Interpretation

To understand the meaning of each cluster, we examined the average CDS spread curves (PX1–PX10) for companies within each group. We computed the mean spread at each tenor within each cluster and visualized the average term structures.

4.3.1 Average CDS Curves by Cluster

```
# Code to summarize and plot average PX1-PX10 curves by cluster

#### Add Cluster Labels Back to the Data
cds_clustered <- cbind(cds_avg, Cluster = pca_df$Cluster)
```

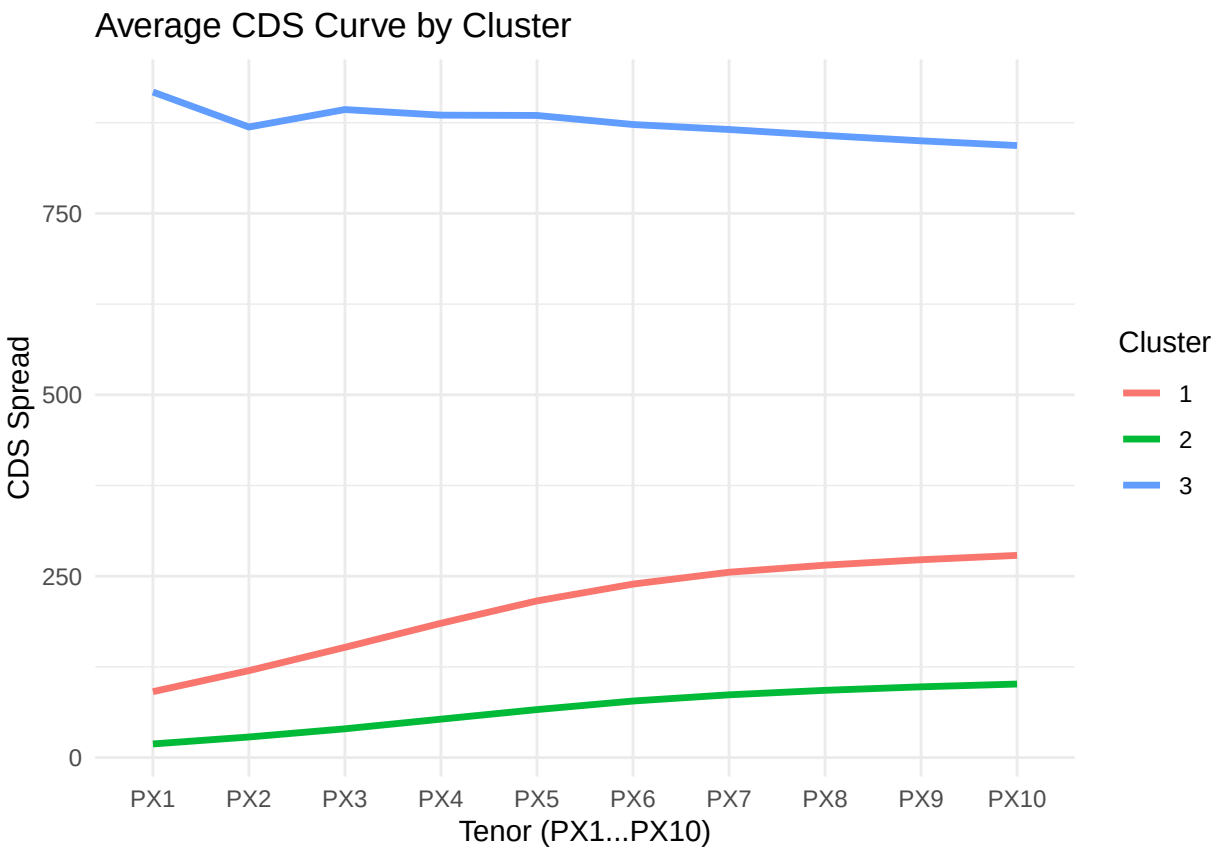
4.3.2 Summarize CDS Spreads by Cluster

```
library(dplyr)
cluster_summary <- cds_clustered %>%
  group_by(Cluster) %>%
  summarise(across(starts_with("PX"), mean, .names = "avg_{.col}"))
library(kableExtra)

cluster_summary %>%
  kbl(caption = "Average CDS Spreads by Cluster (PX1-PX10)", booktabs = TRUE) %>%
  kable_styling(latex_options = c("striped", "hold_position", "scale_down"),
                position = "center")
```

Table 3: Average CDS Spreads by Cluster (PX1–PX10)

Cluster	avg_PX1	avg_PX2	avg_PX3	avg_PX4	avg_PX5	avg_PX6	avg_PX7	avg_PX8	avg_PX9	avg_PX10
1	90.82405	119.8224	151.92883	185.09580	215.95377	239.07582	255.46265	265.16836	272.68128	278.6255
2	18.72972	28.3244	39.63941	52.93295	66.12661	77.93681	86.48463	92.64879	97.47553	101.3432
3	917.34221	869.0669	893.15814	885.43498	885.04111	872.51982	865.66382	857.45357	850.04213	843.5346

```
library(reshape2)
library(ggplot2)
cluster_long <- melt(cds_clustered, id.vars = "Cluster", measure.vars = paste0("PX", 1:10))

ggplot(cluster_long, aes(x = variable, y = value, group = Cluster, color = Cluster)) +
  stat_summary(fun = mean, geom = "line", linewidth = 1.2) +
  labs(title = "Average CDS Curve by Cluster",
       x = "Tenor (PX1-PX10)", y = "CDS Spread") +
  theme_minimal()
```



The average CDS curves show clear differences between clusters:

- Cluster 1: Moderate upward slope — medium risk firms
- Cluster 2: Flat, low spreads — low risk firms
- Cluster 3: High spreads across maturities — high risk or distressed firms

This confirms the clustering reflects real credit differences in credit profiles across firms in terms of level and shape of the CDS term structure. This analysis motivates a deeper look at the relationships between CDS tenors, which we explore next using factor analysis.

4.4 Factor Analysis

To further investigate the structure of the CDS spread data, we applied factor analysis (FA) to the PX1–PX10 variables. This method identifies unobserved latent variables ("factors") that explain correlations among the observed tenors. It provides a complementary view to PCA by focusing on the dependency structure rather than maximizing variance.

We fit a 2-factor model with varimax rotation to improve interpretability.

4.4.1 Select PX Variables and Standardize

```
cds_matrix <- cds_avg[, paste0("PX", 1:10)]
cds_scaled <- scale(cds_matrix)
```

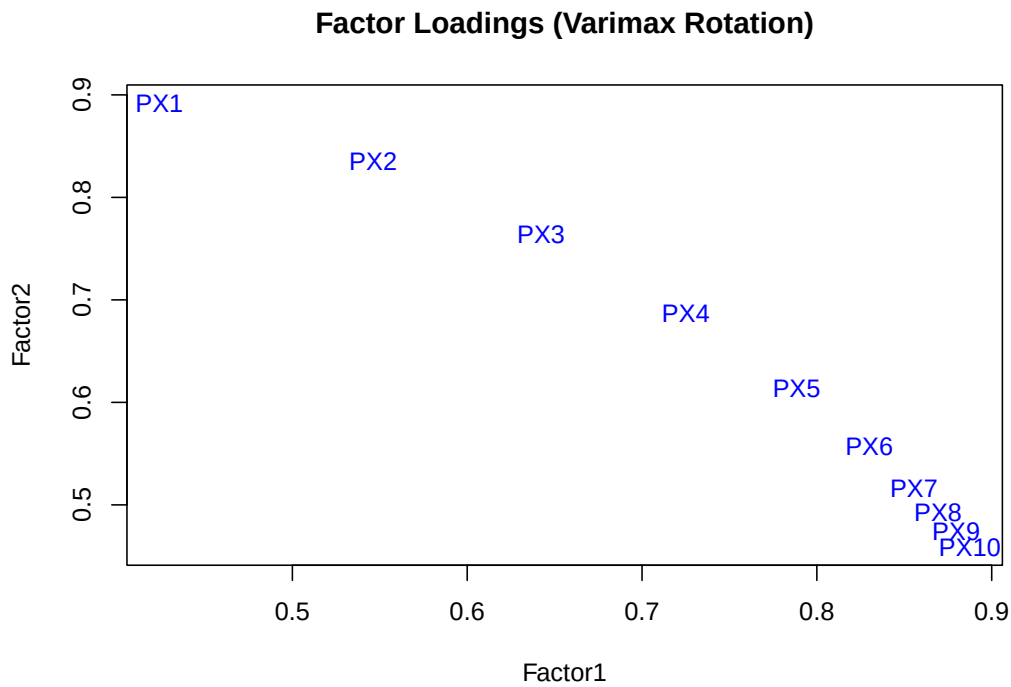
4.4.2 Run Factor Analysis with 2 factors

```
fa_result <- factanal(cds_scaled, factors = 2, rotation = "varimax")
print(fa_result)
```

```
##
## Call:
## factanal(x = cds_scaled, factors = 2, rotation = "varimax")
##
## Uniquenesses:
##   PX1   PX2   PX3   PX4   PX5   PX6   PX7   PX8   PX9  PX10
## 0.024 0.005 0.005 0.005 0.005 0.005 0.005 0.005 0.005 0.005
##
## Loadings:
##      Factor1 Factor2
## PX1  0.424   0.892
## PX2  0.546   0.836
## PX3  0.642   0.765
## PX4  0.725   0.687
## PX5  0.788   0.614
## PX6  0.830   0.557
## PX7  0.856   0.517
## PX8  0.869   0.493
## PX9  0.880   0.474
## PX10 0.888   0.458
##
##                Factor1 Factor2
## SS loadings      5.776   4.183
## Proportion Var   0.578   0.418
## Cumulative Var   0.578   0.996
##
## Test of the hypothesis that 2 factors are sufficient.
## The chi square statistic is 17864.26 on 26 degrees of freedom.
## The p-value is 0
```

4.4.3 Visualize Loadings

```r
# Code to plot factor loadings and label PX1-PX10
loadings <- fa_result$loadings[, 1:2]
plot(loadings, type = "n", main = "Factor Loadings (Varimax Rotation)")
text(loadings, labels = rownames(loadings), col = "blue")
abline(h = 0, v = 0, col = "gray", lty = 2)
```



The factor loading plot shows:

- Factor 1: Driven by PX6–PX10 → long-term credit risk
- Factor 2: Driven by PX1–PX3 → short-term market sentiment

This confirms that the CDS curve structure is driven by at least two interpretable latent factors — one affecting long maturities and another influencing short-term spreads. The results are consistent with financial theory and match the latent groupings we observed in PCA and clustering.

4.5 Interpretation Summary

The results from PCA, clustering, and factor analysis address our research questions directly. We showed that a small number of components (PC1 and PC2) explain nearly all the variation in CDS spreads, satisfying our first question. K-means clustering identified three distinct groups of companies with different credit risk profiles, answering the second. Finally, factor analysis confirmed that the CDS curve is shaped by two interpretable latent factors — short- and long-term risk — supporting our third question.

These findings were consistent with our expectations based on financial theory, though the dominance of a single principal component (PC1 explaining over 95% of the variance) was stronger than anticipated. Overall, the methods worked well together to reveal clear multivariate structure in the data.

5 Conclusion

This project used PCA, clustering, and factor analysis to explore the structure of CDS spreads across companies. We found that most variation in spreads can be captured by one or two dimensions, reflecting shared risk dynamics across firms. Clustering revealed three distinct groups of companies—low, moderate, and high risk—based on their term structures. Factor analysis supported these findings, uncovering two latent factors linked to short- and long-term credit risk. Together, these methods provided a clear multivariate view of how firms differ in market-perceived credit risk.